Reaction-wheel comparison for spiral-outward assembly

Goal: hold structure attitude at zero (exact cancellation). The wheel must store h_w = -h_rel and apply torque -dh_rel/dt.

What the wheels must do at full robot speed (analytic, 0.01 s grid)

Quantity	x	y	z	Unit
Momentum, peak per axis (x / y / z)	3.51	4.01	6.90	N m s
Torque, peak per axis (x / y / z)	12.85	14.03	25.55	N m
Peak momentum, vector norm	7.58	at t = 242 s		N m s
Peak torque, vector norm	28.21	at t = 242 s		N m
Final wheel momentum	-0.0e+00	8.3e-17	1.0e-16	N m s
Nominal robot motion time	264 s	4.40 min		

Headline

- Full-speed need is far above any flown wheel's torque. Fix: slow each robot move until the wheel keeps up (cost: assembly time).
- Torque, not momentum, limits the mission. RW8 (more momentum) gives the same time as RW4 (same 0.25 N m).
- 4RW0 (CubeSat unit) needs 234 min of motion per-move (1212 min uniform).
- BCT RW4-HT x3 (recommended in ZO-25): 13.1 min per-move, 35.2 min uniform, 9.6 kg.
- Open: wheel power at full speed (5.1 kW real peak) is why slowing is needed; wheel mass (up to 9.6 kg) is not yet in the inertia.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Candidate wheel sets (vendor datasheet values)

Per-axis limits. BCT sets are three wheels, one per body axis. 4RW0 is a 4-wheel pyramid; per-axis values are as given on its product page.

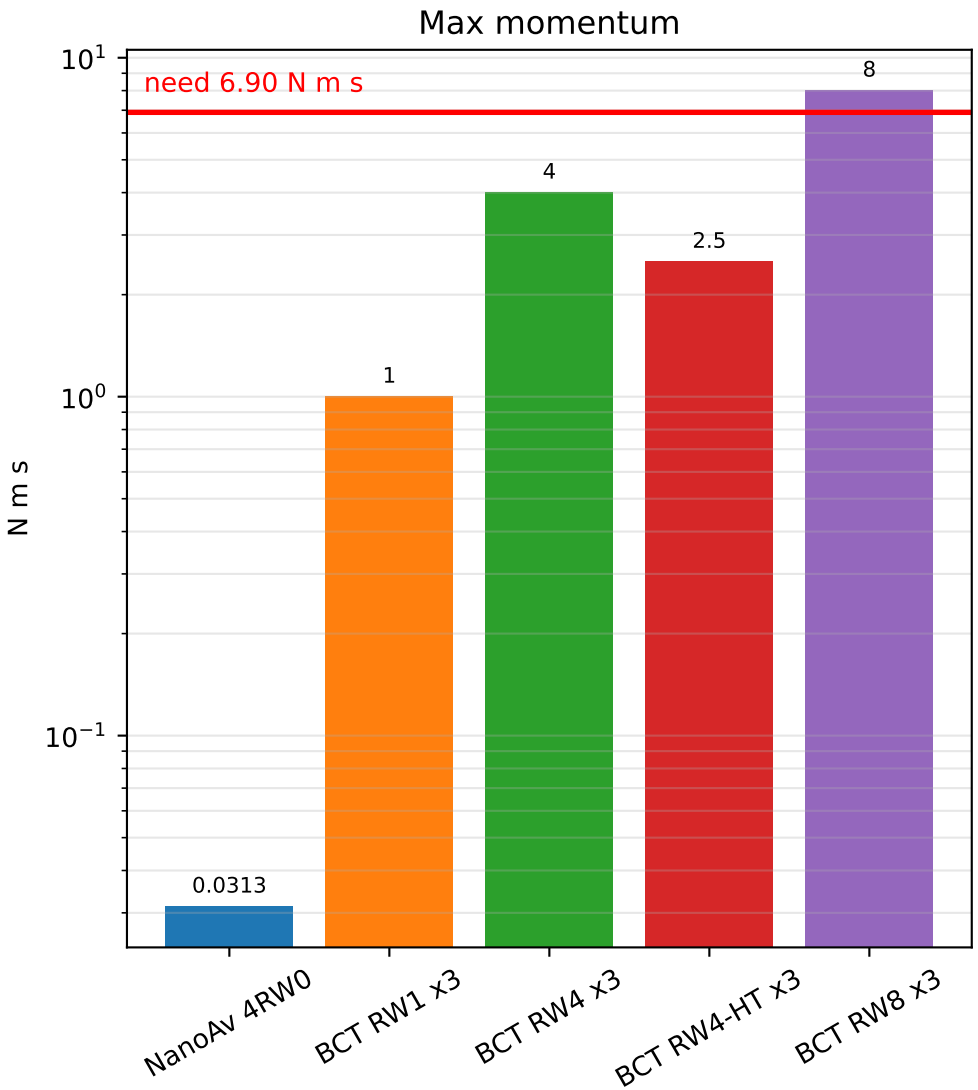
Wheel set	H max x/y/z (N m s)	T max x/y/z (N m)	Mass (kg)	Peak power (W, total)	Heritage
NanoAv 4RW0	0.037/0.037/0.0313	0.0059/0.0059/0.005	0.67	n/a	CubeSat-class unit (vendor page; flight count not stated in screenshots)
BCT RW1 x3	1/1/1	0.06/0.06/0.06	3.30	300	RW1: 162 units launched, 208 cumulative years, longest 3.3 years
BCT RW4 x3	4/4/4	0.25/0.25/0.25	9.60	528	RW4 family: 94 units launched, 190 cumulative years on orbit, longest unit 5.2 years
BCT RW4-HT x3	2.5/2.5/2.5	0.5/0.5/0.5	9.60	528	RW4 family: 94 units launched, 190 cumulative years on orbit, longest unit 5.2 years
BCT RW8 x3	8/8/8	0.25/0.25/0.25	13.20	528	RW8 family: 14 units launched, 44 cumulative years, longest 5.2 years
NEED at full speed	3.51/4.01/6.90	12.85/14.03/25.55	-	5100 (real peak)	ZO-25 analytic requirement

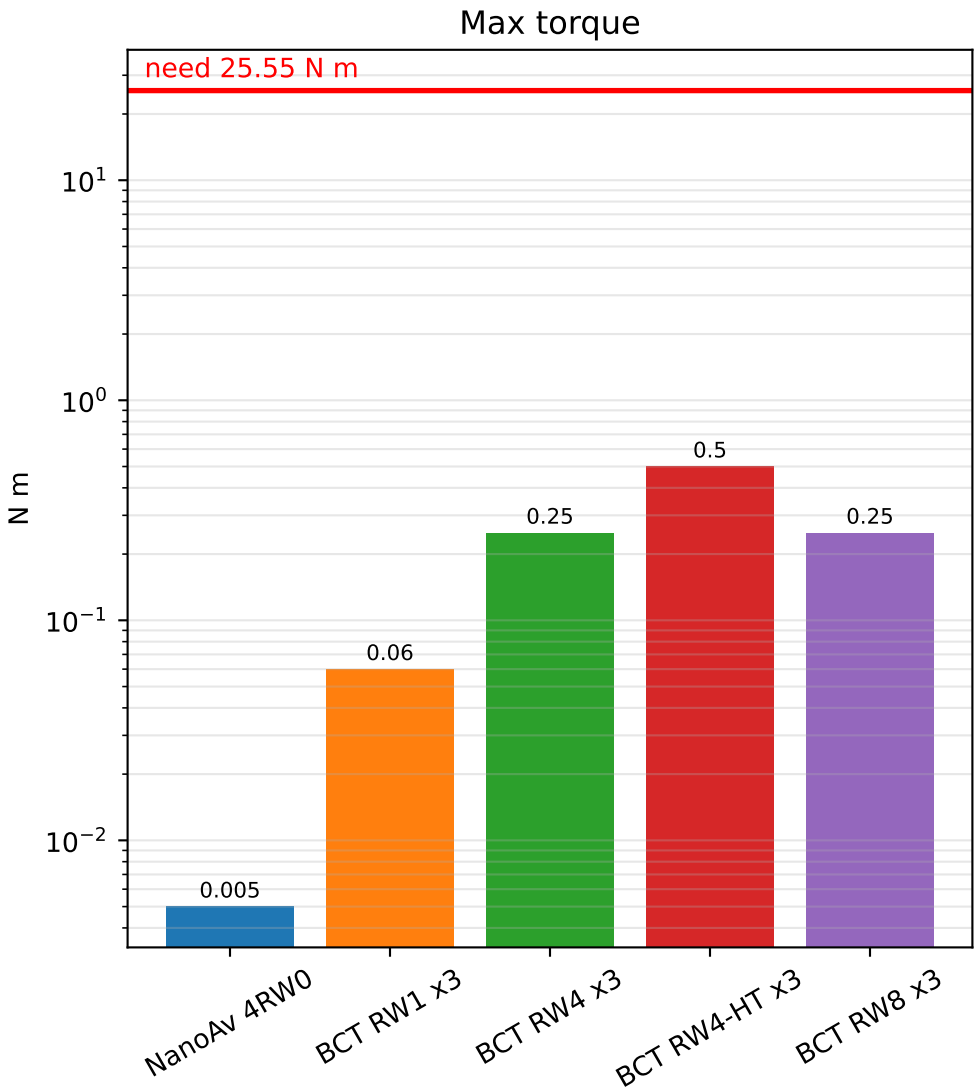
Sources: NanoAvionics 4RW0 product page (user screenshots in 'Reaction Wheel/'); Blue Canyon Technologies, 'Reaction Wheels Full Product Line', REV 1/2026. RW4-HT trades momentum (2.5 vs 4.0 N m s) for double torque (0.50 vs 0.25 N m). BCT heritage numbers are for the RW4 family; the datasheet does not split RW4 and RW4-HT.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Capability against need (z axis, the worst axis)

Bars are wheel limits. Red line is the full-speed need. Anything below the red line forces the robot to slow down.



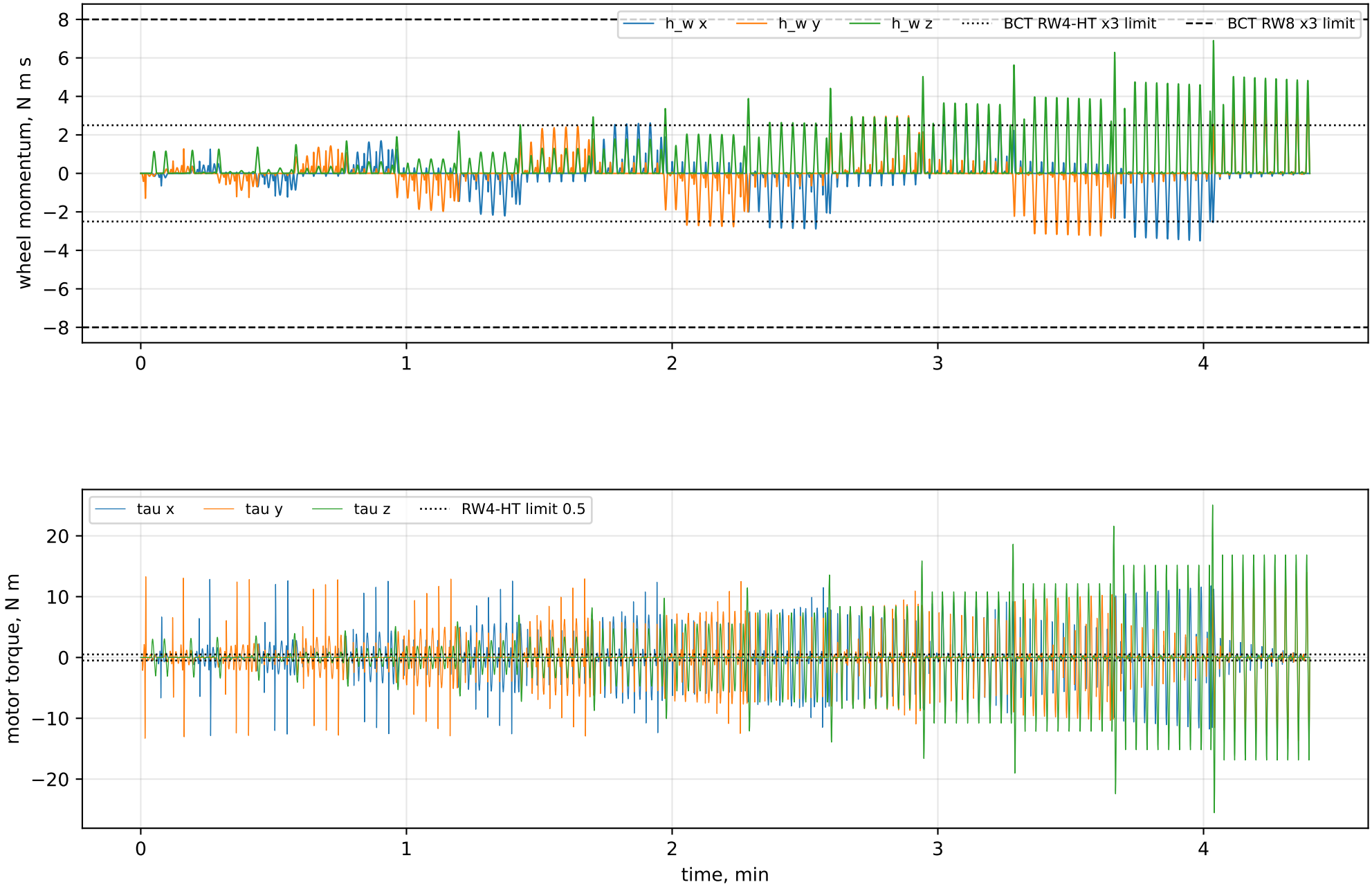


Torque shortfall is the larger gap: even the best set (0.50 N m) is ~51x below the 25.55 N m peak. Momentum shortfall is ~2.8x for RW4-HT and absent only for RW8 (8 N m s > 6.90). Slowing a move by s cuts momentum need by s and torque need by s^2.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

What the spiral assembly asks of the wheels (full speed)

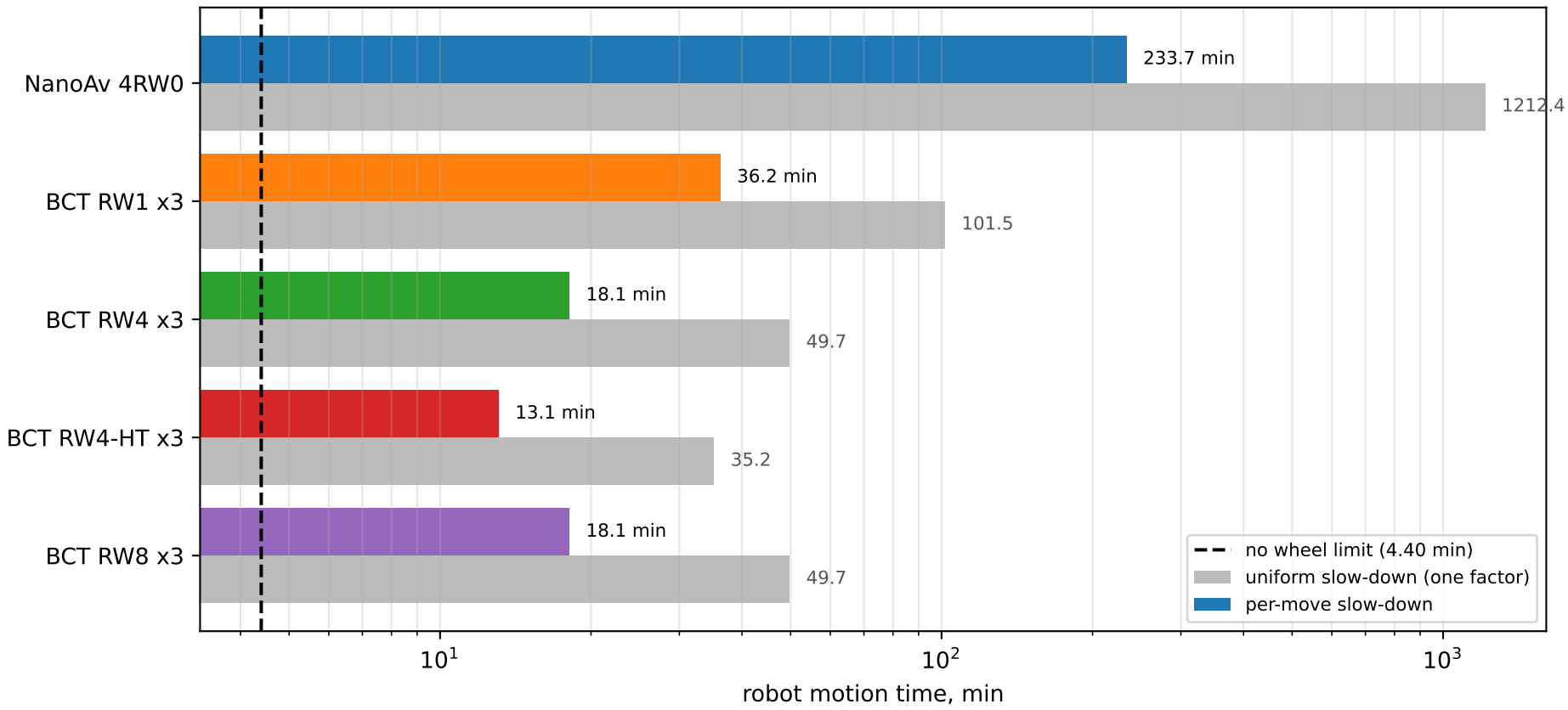
Wheel momentum h_w = -h_rel and motor torque -dh_rel/dt, per body axis. Dotted lines: limits of two candidate sets.



Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Cost of using each wheel: robot motion time

Robot moves are slowed (same path) until the wheel stays within 80% of its momentum and torque limits. Log scale.

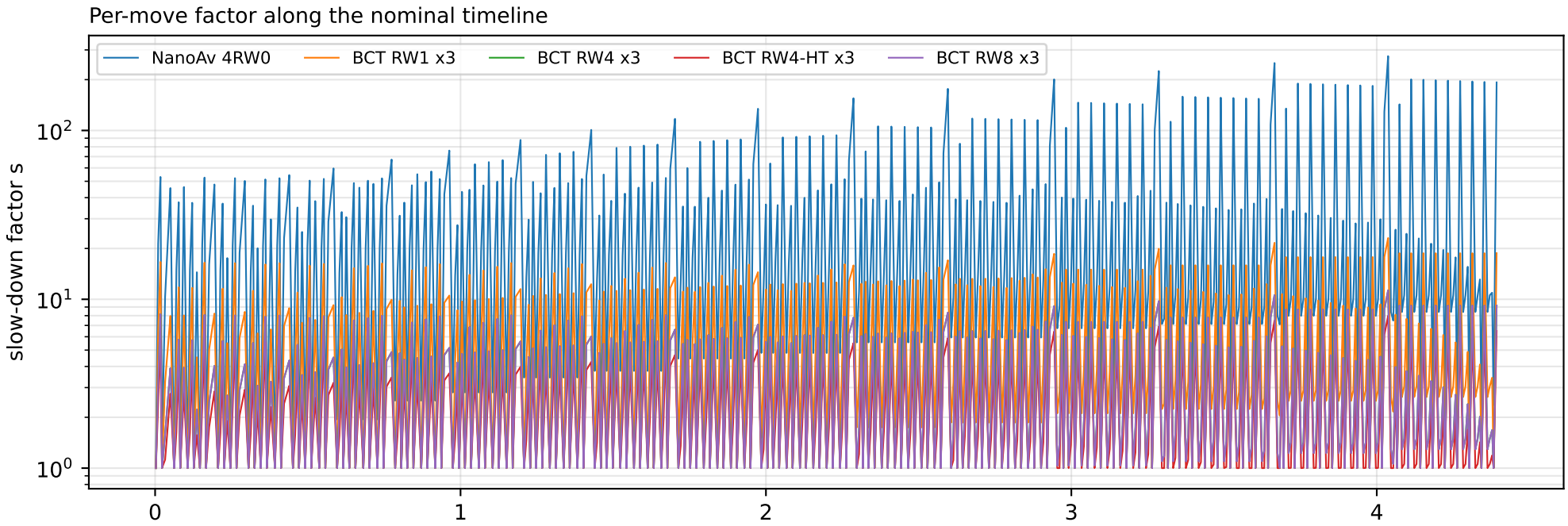


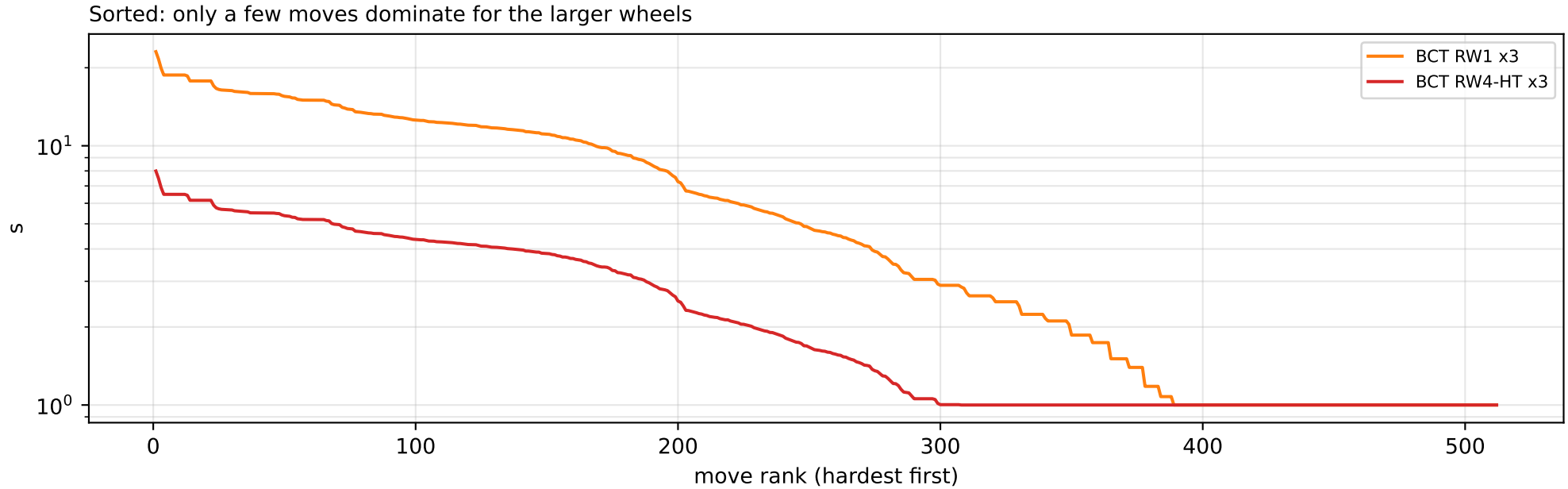
Wheel set	Per-move (min)	Uniform (min)	Per-move vs nominal	Worst move slow-down s	Moves limited by torque / momentum / no	Mass (kg)
NanoAv 4RW0	233.7	1212.4	53.1x	275.6	386 / 110 / 16	0.7
BCT RW1 x3	36.2	101.5	8.2x	23.1	388 / 0 / 124	3.3
BCT RW4 x3	18.1	49.7	4.1x	11.3	349 / 0 / 163	9.6
BCT RW4-HT x3	13.1	35.2	3.0x	8.0	307 / 0 / 205	9.6
BCT RW8 x3	18.1	49.7	4.1x	11.3	349 / 0 / 163	13.2

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Where the time goes: slow-down factor for each of the robot moves

512 moves between rest states. Factor s = 1 means the move runs at nominal speed.

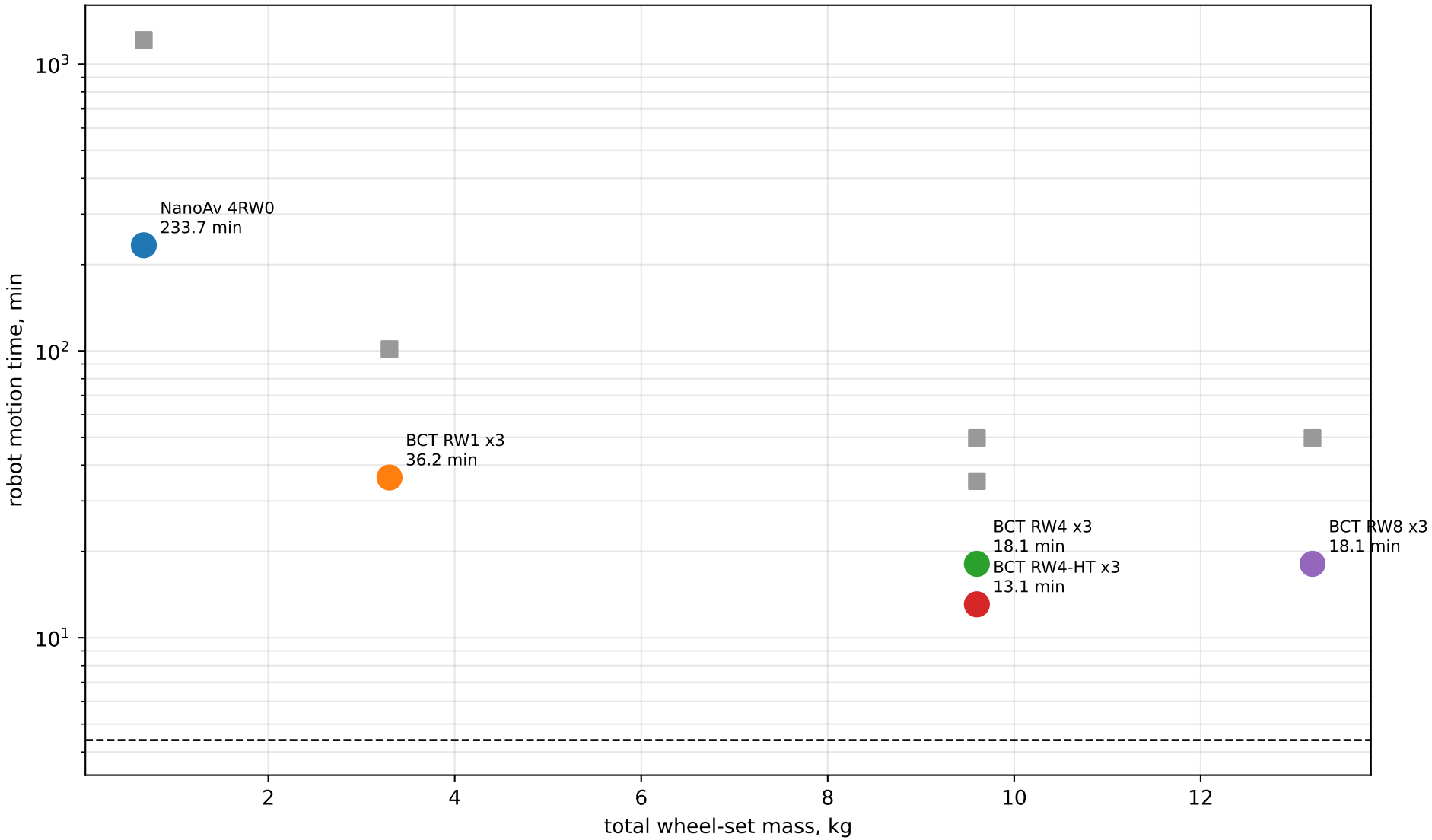




Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Trade: motion time against wheel mass

Marker label = per-move motion time. Lower left is better. Uniform times are the grey markers.

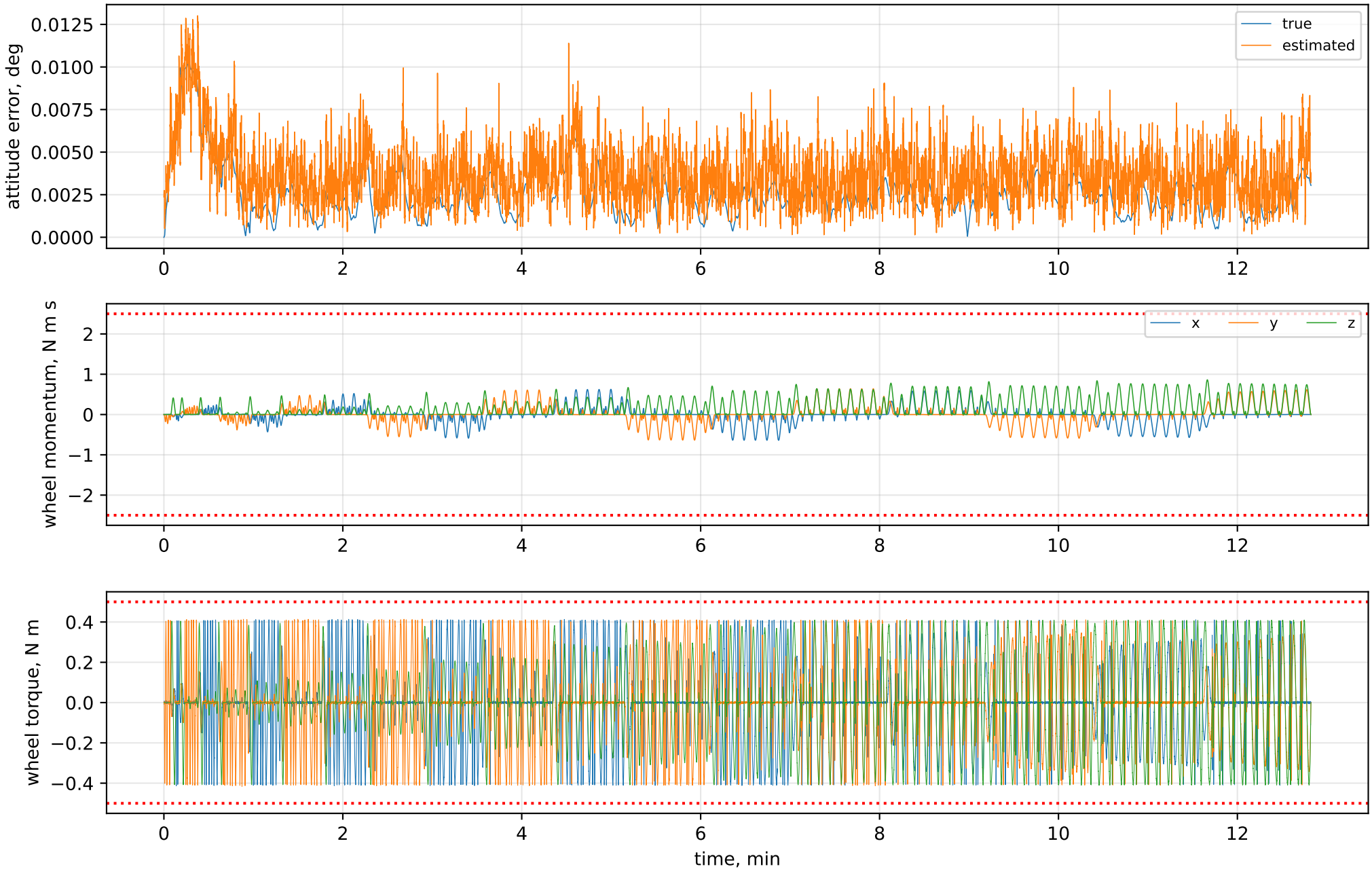


Heavier wheels buy back time, but the buy stops when torque saturates (RW8 = RW4 in time, 1.2 kg heavier per axis).
Wheel power (176 W peak per BCT RW4/RW8 wheel; 100 W RW1) must come from the structure power system.

Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Closed-loop check: BCT RW4-HT x3 (high torque)

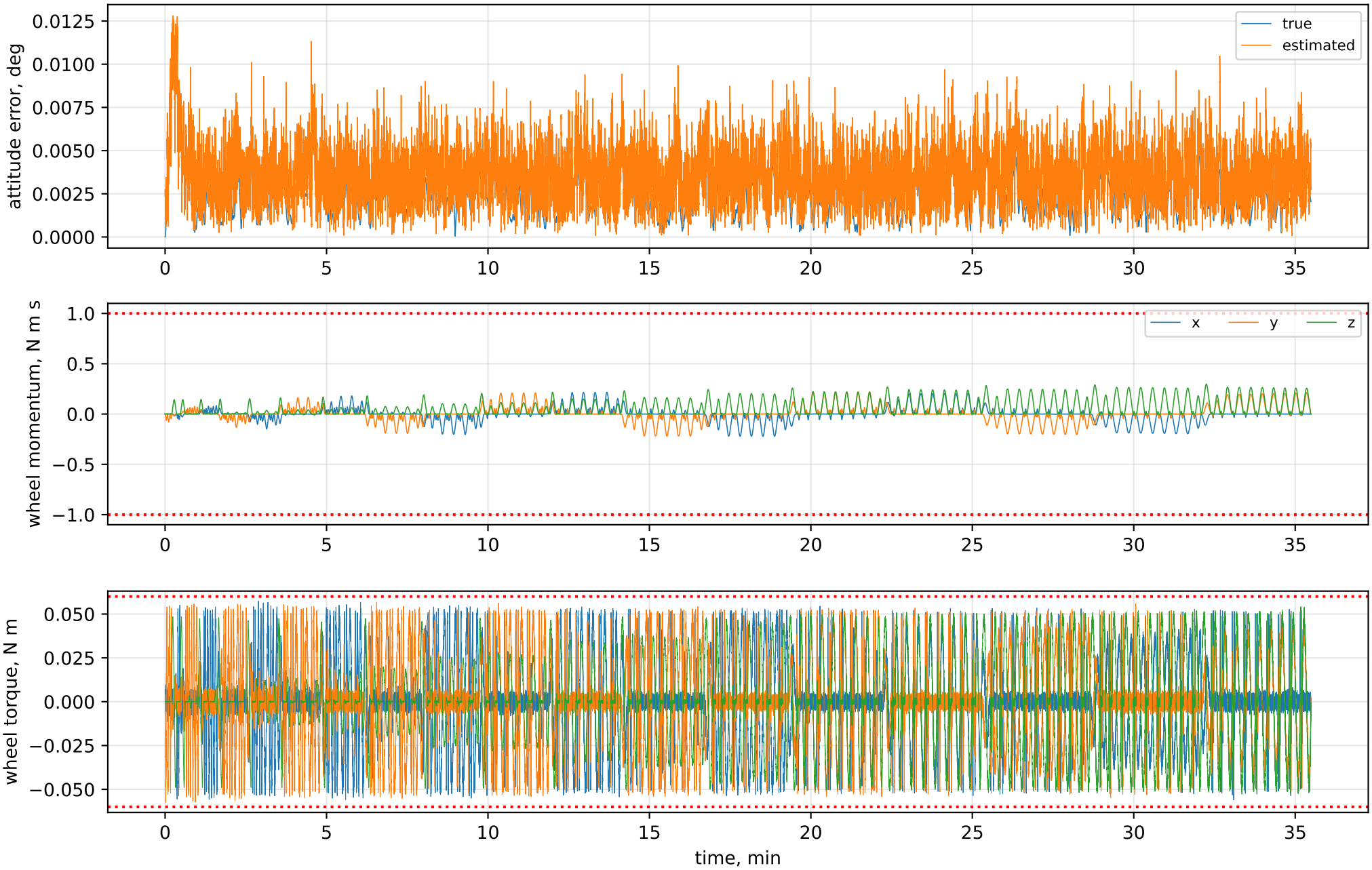
Flight-style controller, noisy gyro and star tracker, 1-cycle report delay, robot 2% faster than plan. Motion 13.1 min. Peak true attitude error 0.0104 deg, final 0.0030 deg. Limit hits: {'sat_torque': 0, 'sat_momentum': 0}.



Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).

Closed-loop check: BCT RW1 x3

Flight-style controller, noisy gyro and star tracker, 1-cycle report delay, robot 2% faster than plan. Motion 36.2 min. Peak true attitude error 0.0101 deg, final 0.0020 deg. Limit hits: {'sat_torque': 0, 'sat_momentum': 0}.



Rigid, isolated, prescribed-motion model. Wheel mass not added to structure inertia. Spiral-outward assembly. Source: ZO-25 (wheel_sizing.py, wheel_control.py).